



## Motivation

**Classical Galerkin schemes** (finite elements, discontinuous Galerkin):

- Trial/test functions are piecewise polynomials;
- Discrete spaces applicable to a wide range of PDEs;
- Good accuracy may require large linear systems

**Goal:** To design **high-order** numerical methods with **reduced computational cost**, using discrete spaces that are tailored to the PDE itself.

**Fields:** Wave problems in acoustics, electromagnetics, etc.

## Trefftz and Quasi-Trefftz Methods

**Trefftz methods.** High-order discontinuous Galerkin schemes whose trial and test functions are **exact** solutions of the PDE on every mesh element  $T \in \mathcal{T}_h$ .

*Example:* Plane waves  $e^{i\omega \mathbf{d} \cdot \mathbf{x}}$  for Helmholtz or Maxwell's equations.

✓ Accuracy with reduced computational cost for linear, homogeneous PDEs with piecewise-*constant* coefficients

✗ *Not* applicable for *variable* coefficients (e.g., wave propagation in inhomogeneous media).

⇒ **Quasi-Trefftz methods.** Trial/test functions are only element-wise **approximate** solutions, preserving the favorable properties of Trefftz schemes while gaining flexibility for **variable coefficients**.

## Quasi-Trefftz Spaces: Scalar Problems

$$\mathcal{L}u = 0 \quad \text{in } \Omega \subset \mathbb{R}^d$$

where  $\mathcal{L}$  is a linear differential operator of order  $\gamma$  with smooth coefficients.

**Local quasi-Trefftz space** on element  $T$ , centered at  $\mathbf{x}_T \in T$ :

$$\mathbb{Q}\mathbb{T}_p(T) = \{ Q \in \mathbb{P}_p : \mathcal{T}_{p-\gamma}^{\mathbf{x}_T} \circ \mathcal{L} Q = 0 \} \subset \mathbb{P}_p$$

$\mathcal{T}_q^{\mathbf{x}_T}$ : Taylor operator of order  $q$  at  $\mathbf{x}_T$

Functions in  $\mathbb{Q}\mathbb{T}_p(T)$  solve the PDE up to a residual  $O(\|\mathbf{x} - \mathbf{x}_T\|^{p-\gamma+1})$ .

**Main results (scalar case):**

- Efficient *layered* construction algorithm, from the decomposition  $\mathbb{P}_p = \bigoplus_{k=0}^p \tilde{\mathbb{P}}_k$  and surjectivity of the highest-order part of  $\mathcal{L}$  with frozen coefficients;
- Explicit bases, dimension  $O(p^{d-1})$  instead of  $O(p^d)$ ;
- Optimal-order **best-approximation** estimates.

## Extension to Time-Harmonic Maxwell's Equations

Fields  $\mathbf{E}, \mathbf{B}$ , wavenumber  $\kappa > 0$ , variable scalar coefficients  $\varepsilon(\mathbf{x}), \mu(\mathbf{x})$ :

$$\begin{cases} \nabla \times \mathbf{E} - i\kappa \mu \mathbf{B} = \mathbf{0} & \text{in } \Omega \\ \nabla \times \mathbf{B} + i\kappa \varepsilon \mathbf{E} = \mathbf{0} & \text{in } \Omega \end{cases}$$

**Quasi-Trefftz space** for the Maxwell system, on  $T$  centered at  $\mathbf{x}_T$ :

$$\mathbb{Q}\mathbb{T}_p(T) := \left\{ (Q^E, Q^B) \in (\mathbb{P}_p)^3 \times (\mathbb{P}_p)^3 : \begin{cases} \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \times Q^E - i\kappa \mu Q^B) = \mathbf{0} \\ \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \times Q^B + i\kappa \varepsilon Q^E) = \mathbf{0} \\ \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \cdot (\mu Q^B)) = 0 \\ \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \cdot (\varepsilon Q^E)) = 0 \end{cases} \right\}$$

## Helmholtz-Type Decomposition

**Key difficulty:** Unlike the scalar case, the curl operator is *not* surjective on homogeneous polynomial vector fields  $\tilde{\mathbb{P}}_k^3$ , it is only surjective onto **divergence-free** fields:

$$\text{Range}(\nabla \times |_{(\tilde{\mathbb{P}}_{k+2}^3)}) = \ker(\nabla \cdot |_{(\tilde{\mathbb{P}}_{k+1}^3)})$$

**Helmholtz-type decomposition.** Every homogeneous polynomial vector field splits as

$$(\tilde{\mathbb{P}}_k^3) = \underbrace{\tilde{\mathbb{H}}_k}_{\text{harmonic}} \oplus \underbrace{\tilde{\mathbb{S}}_k^*}_{\text{solenoidal}} \oplus \underbrace{\tilde{\mathbb{I}}_k^*}_{\text{irrotational}}$$

$$\tilde{\mathbb{S}}_k = \ker(\nabla \cdot |_{(\tilde{\mathbb{P}}_{k+1}^3)}), \tilde{\mathbb{I}}_k = \ker(\nabla \times |_{(\tilde{\mathbb{P}}_{k+2}^3)}), \tilde{\mathbb{H}}_k = \tilde{\mathbb{S}}_k \cap \tilde{\mathbb{I}}_k, \tilde{\mathbb{S}}_k = \tilde{\mathbb{S}}_k^* \oplus \tilde{\mathbb{H}}_k, \tilde{\mathbb{I}}_k = \tilde{\mathbb{I}}_k^* \oplus \tilde{\mathbb{H}}_k.$$

A quasi-Trefftz function is built **layer by layer**, degree  $k = 1, \dots, p$ , solving small linear problems for the parts solenoidal  $\mathbf{F}_k^*$ , irrotational  $\mathbf{G}_k^*$ , given harmonic components  $\mathbf{H}_k^*$ :

$$\mathbf{Q}^* = \mathbf{H}_0^* + \sum_{k=1}^p (\mathbf{H}_k^* + \mathbf{F}_k^* + \mathbf{G}_k^*) \quad \star = E, B$$

## Layered Construction Algorithm

**(Input)**  $p \in \mathbb{N}$ ,  $\mathbf{x}_T \in T$ ,  $\varepsilon_k, \mu_k \in \tilde{\mathbb{P}}_k$ ,  $\mathbf{H}_k^E, \mathbf{H}_k^B \in \tilde{\mathbb{H}}_k$ ,  $k = 0, \dots, p$

**(Step**  $k = 1, \dots, p$ )

▷ Find  $\mathbf{F}_k^E, \mathbf{F}_k^B \in \tilde{\mathbb{S}}_k^*$  such that

$$\begin{cases} \nabla \times \mathbf{F}_k^E = i\kappa \left( \mu_{k-1} \mathbf{H}_0^B + \sum_{j=1}^{k-1} \mu_{k-1-j} (\mathbf{F}_j^B + \mathbf{G}_j^B + \mathbf{H}_j^B) \right) \\ \nabla \times \mathbf{F}_k^B = -i\kappa \left( \varepsilon_{k-1} \mathbf{H}_0^E + \sum_{j=1}^{k-1} \varepsilon_{k-1-j} (\mathbf{F}_j^E + \mathbf{G}_j^E + \mathbf{H}_j^E) \right) \end{cases}$$

▷ Find  $\mathbf{G}_k^E, \mathbf{G}_k^B \in \tilde{\mathbb{I}}_k^*$  such that

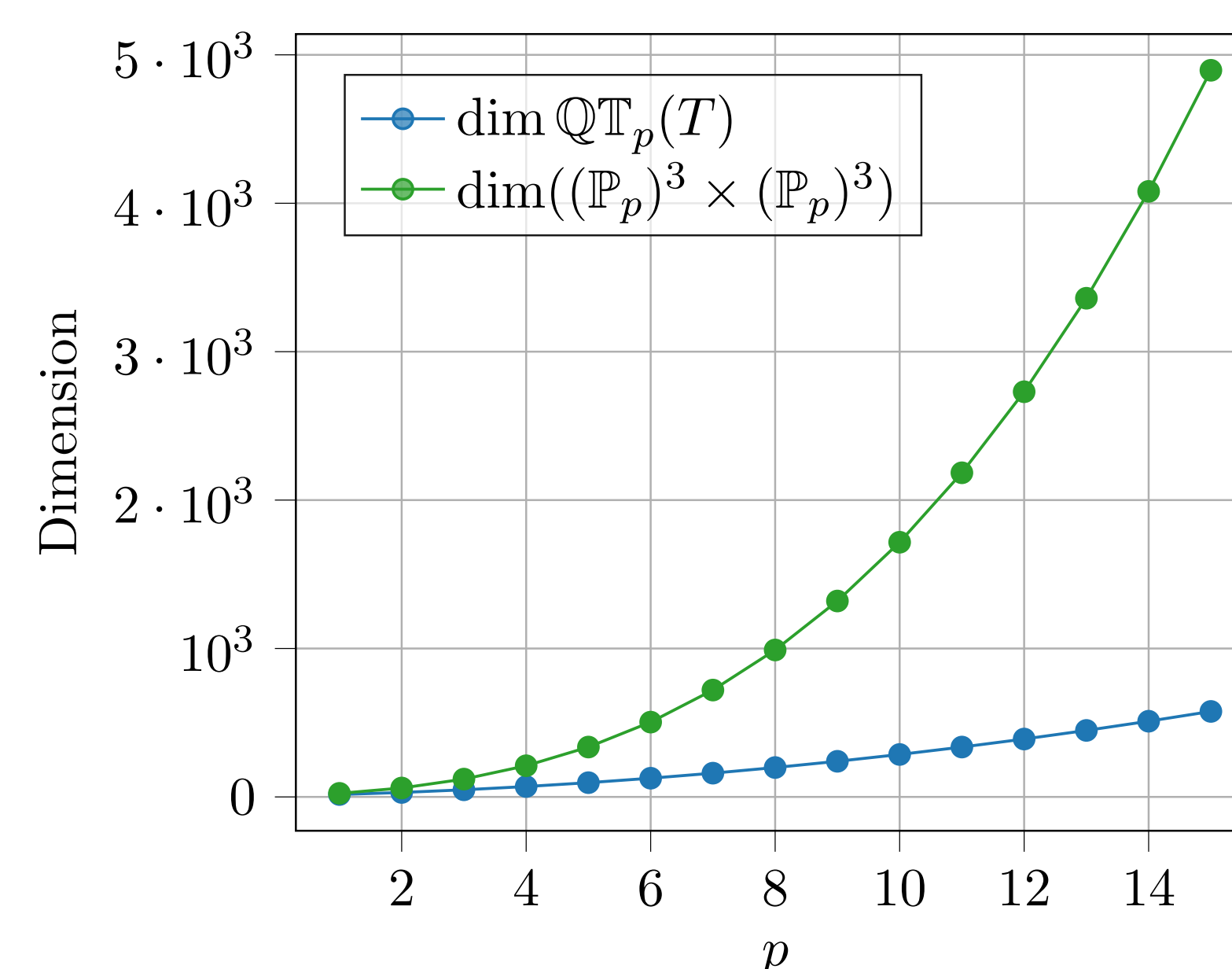
$$\begin{cases} \nabla \cdot \mathbf{G}_k^E = -\varepsilon_0^{-1} \nabla \cdot \left( \varepsilon_k \mathbf{H}_0^E + \sum_{j=1}^{k-1} \varepsilon_{k-j} (\mathbf{F}_j^E + \mathbf{G}_j^E + \mathbf{H}_j^E) \right) \\ \nabla \cdot \mathbf{G}_k^B = -\mu_0^{-1} \nabla \cdot \left( \mu_k \mathbf{H}_0^B + \sum_{j=1}^{k-1} \mu_{k-j} (\mathbf{F}_j^B + \mathbf{G}_j^B + \mathbf{H}_j^B) \right) \end{cases}$$

**(Output)**  $(Q^E, Q^B) \in \mathbb{Q}\mathbb{T}_p(T)$ ;  $\mathbf{Q}^* = \mathbf{H}_0^* + \sum_{k=1}^p (\mathbf{H}_k^* + \mathbf{F}_k^* + \mathbf{G}_k^*)$ ,  $\star = E, B$

## Basis and Dimension of the Quasi-Trefftz Space

$$\dim \mathbb{Q}\mathbb{T}_p(T) = \dim(\mathbb{H}_p \times \mathbb{H}_p) = 2(p+1)(p+3) = O(p^2)$$

$$\dim((\mathbb{P}_p)^3 \times (\mathbb{P}_p)^3) = (p+1)(p+2)(p+3) = O(p^3)$$



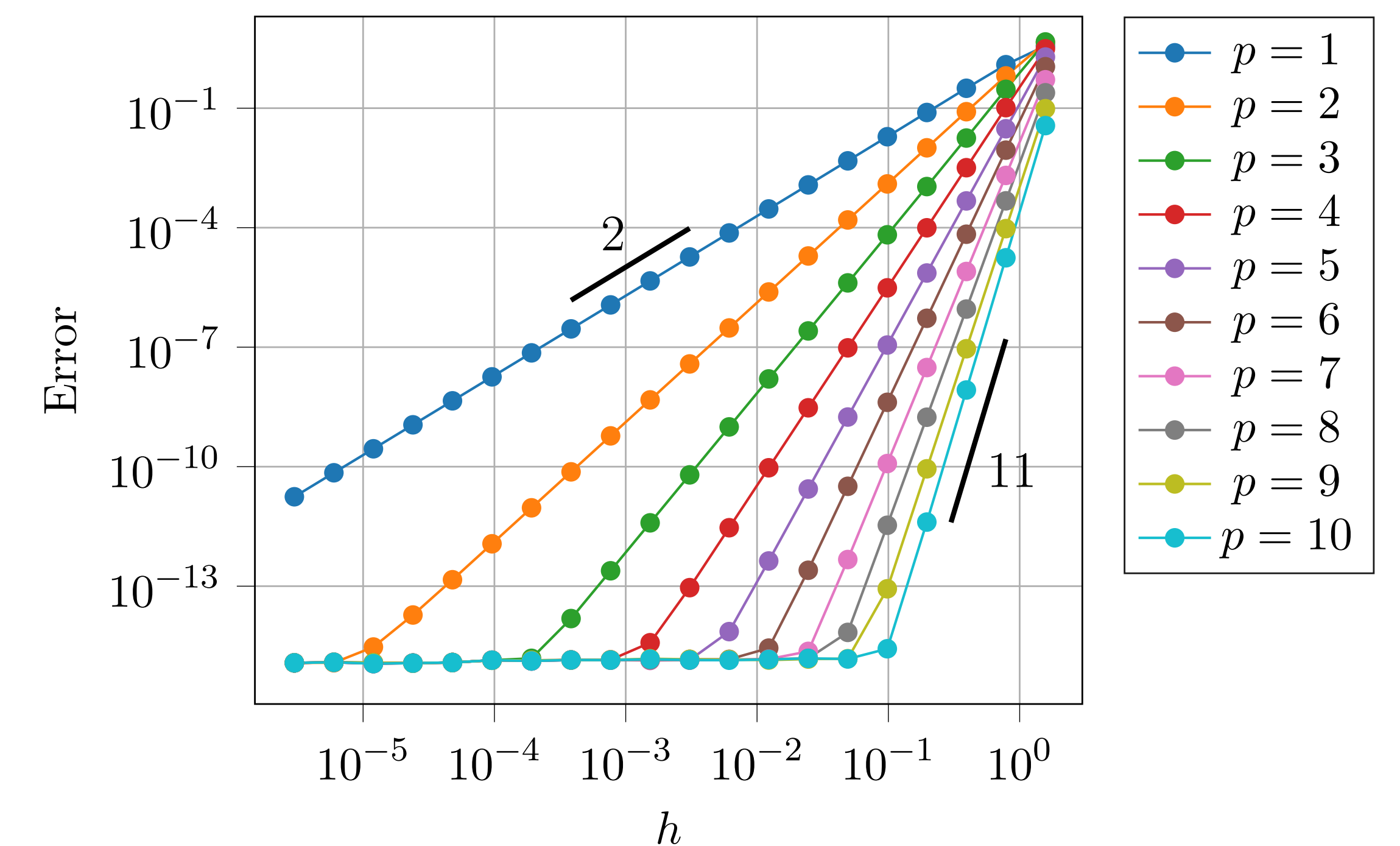
## Approximation Properties & Numerical Validation

**Theorem (Best Approximation Properties).** Given  $p \in \mathbb{N}$  and  $T \subset \mathbb{R}^3$ , suppose  $\varepsilon, \mu \in C^p(T; \mathbb{C})$  and  $(\mathbf{E}, \mathbf{B}) \in (C^{p+1}(T; \mathbb{C}^3))^2$  solves Maxwell's equations in  $T$ . Then,

$$(\mathcal{T}_p^{\mathbf{x}_T} \mathbf{E}, \mathcal{T}_p^{\mathbf{x}_T} \mathbf{B}) \in \mathbb{Q}\mathbb{T}_p(T),$$

and for  $T$  star-shaped w.r.t.  $\mathbf{x}_T$ ,

$$\inf_{(Q^E, Q^B) \in \mathbb{Q}\mathbb{T}_p(T)} (\|\mathbf{E} - Q^E\| + \|\mathbf{B} - Q^B\|) \leq \frac{3^{p+1}}{(p+1)!} h_T^{p+1} (\|\mathbf{E}\| + \|\mathbf{B}\|).$$



**Error:**  $\sup_{\|\mathbf{x} - \mathbf{x}_T\| \leq h} (\|\mathbf{E}(\mathbf{x}) - Q^E(\mathbf{x})\| + \|\mathbf{B}(\mathbf{x}) - Q^B(\mathbf{x})\|)$

## Conclusions & Perspectives

- Quasi-Trefftz spaces retain the approximation properties of Trefftz methods while accommodating **variable coefficients**, with dimension *smaller* than the full polynomial space.
- The layered construction, developed for scalar PDEs, extend to the **first-order time-harmonic Maxwell system** with some extra tools like the Helmholtz-type decomposition.

**Ongoing and future work:**

- Quasi-Trefftz **DG variational formulation** for Maxwell's equations, and its implementation
- Extension to further PDEs
- Development of *a posteriori* error estimation and adaptive algorithms.

## References

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